Midterm Project Proposal

Medical Multi-Agent LLM System for Factual Accuracy on MedQA-USMLE

1. Project Motivation

Build a medical multi-agent LLM system that improves factual accuracy on MedQA-USMLE. The project evaluates system design rather than clinical deployment.

2. Learning Objectives

Design a modular multi-agent architecture, implement a reproducible evaluation pipeline, compare against a direct LLM baseline, and analyze component contributions.

3. System Task

Input: one MedQA-USMLE multiple-choice question.
Output: exactly one final answer option plus a short explanation.

4. Required System Components

Required: (1) Direct LLM baseline, (2) Medical reasoning agent, (3) RAG module, (4) Short-term memory, (5) Long-term memory, (6) Multi-agent workflow (≥3 agents), (7) Optional evolutionary optimization.

5. Benchmark Protocol

Development: 100–150 public MedQA training/dev questions.
Official evaluation: full MedQA-USMLE test set.
Development data may be used only for debugging and tuning. Official test set is used once for final evaluation.

6. Required System Variants

V0 Direct LLM
V1 RAG-only
V2 Multi-agent without memory
V3 Full system
V4 Full system without verifier (optional)

7. Evaluation Metrics

Primary metric:
Accuracy = (# Correct Predictions) / (Total Number of Questions)

For the official evaluation:

Accuracy = (# Correct Predictions) / 1273

Additional metrics:
- Invalid response rate = (# Invalid Responses) / (Total Number of Questions)
- Accuracy gain = Accuracy(V3) − Accuracy(V0)
- Ablation comparison
- Win/Loss/Tie analysis
- Bootstrap confidence interval and McNemar test (recommended)
- Cost and latency

8. Required Result Tables

Leaderboard table, paired comparison table, and error analysis table.

9. Implementation Requirements

Runnable source code, documented repository structure, prediction files, evaluation script, and report.

10. Reproducibility Requirements

Report model, prompts, temperature, retrieval settings, hardware/API, random seed, and runtime configuration.

12. Midterm Report Structure

1. Introduction
2. System Architecture
3. System Design
   - Agent architecture
   - RAG design
   - Memory design
   - Verifier design
   - Evolutionary optimization (if used)
4. Experimental Setup
   - Model and prompts
   - Retrieval configuration
   - Hardware/API settings
5. Evaluation Results
   - Main accuracy table
   - Baseline comparison
   - Ablation study
   - Win/Loss/Tie analysis
   - Cost and latency
6. Error Analysis
7. Discussion
8. Limitations
9. Conclusion

13. Submission Package

Source code, report PDF, prediction files, prompts, architecture diagram, presentation slides.

17. Presentation Requirements

7–10 minute presentation covering architecture, evaluation, ablations, and lessons learned.

18. Final-Project Connection

The submitted system will be reused as the target system for the final attack/defense project.

19. Minimum Acceptance Criteria

Runnable baseline and full system, official prediction files, evaluation results, at least one ablation, and report.

Verified Equations

Accuracy = Correct Predictions / Total Number of Questions

Official benchmark accuracy = Correct Predictions / Dataset size

Invalid Response Rate = Invalid Responses / Total Number of Questions

Accuracy Gain = Accuracy(V3) − Accuracy(V0)